

ASTRA

AI Desktop Agent — Complete Tech Stack

Runtime & Language

Node.js (v18+)

Primary runtime environment — async, non-blocking, perfect for I/O-heavy automation tasks.

JavaScript / ES Modules

Use modern import/export syntax and async/await throughout the codebase.

Browser Automation

Playwright

Controls Chromium/Firefox/WebKit browsers — clicks, typing, navigation, screenshots. Install: `npm install playwright`

@playwright/test

Optional test runner for verifying browser automation flows during development.

Desktop Control

RobotJS

Native desktop automation — mouse movement, keyboard input, screen capture. Node.js only library. Install: `npm install robotjs`

screenshot-desktop

Cross-platform screen capture to Base64 or file. Lightweight alternative to RobotJS screenshot. Install: `npm install screenshot-desktop`

AI / Vision Layer

Ollama

Local LLM server running on localhost:11434. Runs models like LLaMA 3, Mistral, Phi-3 offline on your machine.

ollama (npm)

Official Node.js SDK for Ollama. Supports chat, streaming, and vision (image input). Install: `npm install ollama`

LLaVA / Llama3.2-Vision

Vision-capable models that can 'see' screenshots and decide what action to take next.

Communication / Coordination

ngrok

Exposes local ASTRA server to the internet so your Next.js scanner app can talk to it remotely.

Express.js	Lightweight HTTP server to receive commands from the scanner frontend. Install: npm install express
WebSocket (ws)	Real-time two-way communication between ASTRA and scanner UI. Install: npm install ws

Frontend (Scanner UI)

Next.js	React framework for your scanner app — already running on port 3000 in your setup.
React	Component-based UI for building the scanner/control dashboard.
Tailwind CSS	Utility-first CSS for fast, clean UI styling without custom CSS files.

Process Management

PM2	Keeps ASTRA running as a background process, auto-restarts on crash. Already in your InvenTree setup.
dotenv	Loads environment variables (API keys, ports, model names) from a .env file. Install: npm install dotenv

Developer Tooling

nodemon	Auto-restarts Node.js on file changes during development. Install: npm install -D nodemon
ESLint + Prettier	Code linting and formatting to keep the codebase clean and consistent.
Git	Version control — track changes, roll back, share code with mentor.

ASTRA Core Loop:

1. Capture	Screenshot via RobotJS or screenshot-desktop → convert to Base64
2. Think	Send Base64 image to Ollama (LLaVA) → model decides next action
3. Act	Execute action via RobotJS (click / type) or Playwright (browser)
4. Repeat	Loop back to step 1 until task is complete

All tools run locally on Windows. No cloud dependency required except ngrok for remote access.